Explainer Studio Pro: How It Works and How to Write Your Script

A guide to the Live Explainer Video studio: what each part does, how a script becomes a finished MP4, and how to write scripts that turn out well.

1. What the program does

Explainer Studio Pro turns a written script into a short video. You paste your text, and the program:

1. Splits the script into **scenes** (one scene per line).
2. Picks a **stock video clip** for each scene (from Pixabay, or built-in fallback clips).
3. Shows **captions** over the clips in a live preview.
4. Lets you add a **voiceover** and **background music**.
5. Lets you adjust timing on a **timeline**.
6. **Exports** everything as one MP4 file, using FFmpeg inside your browser.

The home page gives you two choices:

- **Live Explainer Video**: the studio described in this guide.
- **Animation Explainer Video**: empty for now. It is reserved for a future 2D animation tool.

2. Getting started

Run the app

1. Unzip the project and open a terminal in the folder.
2. Run `npm install` (one time).
3. Run `npm run dev` and open the address it prints (usually `http://localhost:5173`).

Tailwind CSS and Font Awesome load from a CDN in `index.html`, so there is nothing extra to set up for styling.

Add your Pixabay key (recommended)

1. Create a free account at pixabay.com and copy your API key from the API documentation page.
2. In the studio, click **API Keys**, paste the key into *Pixabay API Key*, and click **Save Configuration**.
3. The key is stored in your browser (localStorage). It is not sent anywhere except to Pixabay.

Without a key the studio still works, but every scene uses one of five built-in fallback clips and the Pixabay tab cannot search. The *OpenRouter* field is saved but not used yet.

3. The workflow in six steps

1. **Write or paste your script** in the *Script* tab.
2. Click **Parse & Auto-Match Media**. Scenes are created and, if you saved a Pixabay key, each one is matched to a clip.
3. Open the **Scenes** tab to fix text, durations and clips.
4. Open the **Audio** tab to add a voiceover and optional background music.
5. Use the **preview and timeline** to play it back and adjust timing.
6. Click **Render Export** and wait for the MP4 to download.

Important: clicking *Parse* again rebuilds all scenes from the script text. Any edits you made in the Scenes tab, and any clips you applied, are replaced. Finish your script first, then parse once, then fine-tune.

4. How to write the script

The one rule that matters

Every line becomes one scene. The program splits your text wherever there is a line break. Blank lines are ignored, so you can leave a gap between lines for readability. A long paragraph with no line breaks becomes a single very long scene, so press Enter after each sentence or idea.

Write one idea per scene

Each scene gets one clip and one caption. Keep it to a single thought:

- **Length:** aim for 8 to 20 words per line. Most people speak about 2.5 words per second, so a 12-word line fits the default 5-second scene.
- **Style:** short, spoken sentences. Read each line aloud. If you run out of breath, split it.
- **Captions:** the line is shown on screen exactly as you type it, so punctuation and spelling count.

Put concrete nouns early in the sentence

To choose a clip, the program creates a search phrase from the line:

1. Lowercases the text and removes punctuation.
2. Removes common filler words and every word of three letters or fewer.
3. Keeps the **first two words** that remain.
4. Falls back to "technology" if nothing is left.

So the opening words of each line control which video appears.

Line	Search phrase	Result
"Entire websites are now generated automatically."	entire websites	Weak match
"Websites now publish thousands of articles every minute."	websites publish	Better match
"Solar panels turn sunlight into electricity."	solar panels	Strong match
"It is a very big problem for all of us."	very problem	Poor match

Start lines with a visual subject such as *solar panels*, *busy street*, *server room* or *doctor*. Avoid openings like "It is", "There are" or "So the real question".

If the automatic clip is wrong, use **Search Swap** on the scene, or type your own search in the Pixabay tab (see section 5).

A simple structure that works

Part	Scenes	Purpose
Hook	1	A question or surprising fact that makes people keep watching
Problem	1 to 2	What is wrong or confusing

Part	Scenes	Purpose
Explanation	2 to 4	How it works, one step per line
Example	1 to 2	Something concrete the viewer can picture
Close	1	A takeaway or a call to action

For a 30 to 60 second video, 6 to 10 lines is a good target.

Example script

```
Have you ever wondered where your electricity comes from?

Power stations burn fuel to spin giant turbines.

Solar panels skip the burning and turn sunlight straight into power.

Wind turbines catch moving air and spin a generator.

Cities then send that energy through cables to every home.

So next time you flip a switch, you will know the journey behind it.
```

Each line is one idea, starts with something you can see, and fits in about five seconds.

If you will use a voiceover

Write the script first, record the narration reading it line by line, and keep the lines similar in length. The **Sync Timeline to Audio** button divides the audio length **equally** across all scenes, so lines of very different length will not line up with the speech. If one line is much longer, adjust its duration afterwards in the Scenes tab or by dragging on the timeline.

5. The tabs, one by one

Script tab

Paste your text and click **Parse & Auto-Match Media**. The studio loads with a sample script so you can see it working immediately.

Scenes tab

Each scene card shows:

- **Scene number and duration** in seconds (minimum 1). Changing it moves every later scene.
- **Caption text**, which you can edit directly. Editing the text does not change the clip search phrase.
- **Search Swap**, which opens the Pixabay tab and searches using the scene's original phrase.

Add Scene appends a new scene at the end with the text "New scene".

The card for the scene at the playhead is highlighted.

Pixabay tab

1. Type a search and press Enter or click **Search**.
2. Hover a result to preview it.
3. Click **Apply** to put that clip on the scene currently at the playhead. The tab confirms which scene number was changed.

To change a different scene, first click its block on the timeline (or move the playhead into it), then apply. The *Audio* switch in this tab only explains that Pixabay has no music search. Add music from the Audio tab instead.

Audio tab

- **Voiceover:** upload one narration file (MP3, WAV, M4A, OGG) for the whole video. Use the built-in player to check it, **Sync Timeline to Audio** to fit scene durations to it, and the trash button to remove it.
- **Background music:** upload a music file. It loops for the whole video. The **Music Volume** slider (default 20%) sets how loud it is under the voiceover.

6. Preview and timeline

- **Aspect buttons (16:9, 9:16, 1:1)** change the shape of the preview and of the exported video. Use 16:9 for YouTube, 9:16 for Reels, Shorts and TikTok, and 1:1 for square posts.
- **Play, Stop and Restart** control playback. Stop returns to the start. Restart jumps to the start and keeps playing if it was playing.

- **Scene blocks** are shown in order along the timeline. Click a block to jump to that scene. Press and drag a block left or right to shorten or lengthen it (minimum 1.5 seconds).
- **Playhead** is the vertical line. Drag it to scrub, or use the slider under the timeline.
- **Clock** shows the current time and the total length. Total length is the sum of all scene durations.

Voiceover and music follow the playhead, so scrubbing keeps sound in sync. If a clip is shorter than its scene, it loops.

7. How the program works inside

The data

The studio keeps a list of scenes. Each scene has:

Field	Meaning
`id`	A unique ID used to track the scene's video
`text`	The caption
`duration`	Length in seconds
`videoUrl`	The chosen stock clip
`keywords`	The search phrase made from the text

Start times are never stored. They are calculated by adding up the durations of the scenes before each one, so changing one duration moves everything after it automatically.

From script to scenes

1. Split the script on line breaks and drop blank lines.
2. Create a scene for each line with a 5-second default and a fallback clip.
3. If a Pixabay key exists, search for each scene's phrase one at a time and swap in the first result as it arrives.

Clip handling

For each scene the program keeps a hidden `<video>` element. It first **downloads the clip as a file** and plays that local copy. This matters for two reasons: the picture can be copied to the canvas without browser security errors, and FFmpeg can read the same file at export time. If a website blocks the download (a CORS restriction), the clip still plays in the preview, but it cannot be exported.

The live preview

A canvas repaints the current scene's video many times per second, scaled to fill the frame. Captions are a text layer on top. A playback timer advances the playhead, and the video, voiceover and music elements are nudged to match it whenever they drift more than a fraction of a second.

Export with FFmpeg

Clicking **Render Export** runs this sequence entirely in your browser:

1. **Load FFmpeg.** The first export downloads the FFmpeg engine (about 30 MB). Later exports reuse it.
2. **Prepare each scene.** The clip file is loaded into FFmpeg's memory, and the caption is drawn as a transparent PNG image sized for the output.
3. **Build one filter chain.** Every clip is scaled and cropped to fill the output size, set to 30 frames per second, overlaid with its caption, and trimmed to its scene duration. The scenes are joined in order.
4. **Mix audio.** The voiceover and the music (at your chosen volume) are blended. If you added neither, the video has no sound.
5. **Encode.** The result is encoded as an H.264 video with AAC audio and saved as `explainer-studio-export.mp4`.
6. **Clean up.** Temporary files are removed from memory and the download starts.

Export sizes:

Aspect	Exported size
16:9	854 × 480
9:16	480 × 854
1:1	576 × 576

These sizes keep rendering fast and memory use low. They can be raised in the `ASPECTS` setting in `src/utils/helpers.js`.

Project layout

Path	Role
`src/App.jsx`	Switches between home, live studio and animation page
`src/pages/HomePage.jsx`	The two-choice start screen
`src/pages/LiveExplainerPage.jsx`	Holds the studio state and connects all parts
`src/pages/AnimationExplainerPage.jsx`	Empty placeholder
`src/components/live/`	Header, sidebar, tabs, preview, timeline, settings and export overlay
`src/hooks/usePlayback.js`	Playhead timer and play, stop and seek
`src/hooks/useVideoPool.js`	One video element per scene
`src/utils/helpers.js`	Keyword extraction, timing helpers, sizes, clip download
`src/utils/exportVideo.js`	The FFmpeg export pipeline

8. Tips for better videos

- Use Pixabay clips rather than the built-in fallbacks. They download more reliably and match your topic.
- Keep captions to two lines or fewer on screen. Shorter lines look better, especially in 9:16.

- Choose the aspect ratio **before** exporting. Changing it afterwards means rendering again.
- Keep total length reasonable. Long videos use more browser memory. For a long project, export in parts.
- Use music at 10 to 20% so the voice stays clear.
- Preview the whole video once before exporting, since rendering takes longer than playing.

9. Troubleshooting

Problem	Likely cause and fix
Every scene shows the same kind of clip	No Pixabay key saved. Add it under **API Keys**, then parse again.
A clip does not match the line	The first two meaningful words were too vague. Rewrite the start of the line, or use Search Swap or the Pixabay tab.
Pixabay tab says "No API key set" or "Search failed"	Check the key, and check your internet connection.
Preview is black for a scene	The clip is still loading, or the site blocked it. Apply a Pixabay clip instead.
Export says clips were replaced with a black frame	Those clips were blocked by CORS. Replace them with Pixabay clips and export again.
Export fails or the tab runs out of memory	Lower the number or length of scenes, close other tabs, or export a shorter video.
Export stays at "Loading FFmpeg"	The engine could not download. Check your connection and that `unpkg.com` is reachable.
Voiceover ends before the video	Run **Sync Timeline to Audio**, or lengthen the narration.
Voiceover is longer than the video	Audio is cut at the end of the last scene. Increase scene durations.

Problem	Likely cause and fix
My scene edits disappeared	Parse rebuilds all scenes. Avoid clicking it after you start editing.
No music option in Pixabay	Pixabay has no music API. Upload a music file in the Audio tab.

10. Current limits

- The Animation Explainer Video page is empty on purpose.
- The OpenRouter key is stored but nothing uses it yet.
- Automatic clip matching uses only the first two meaningful words of each line.
- Exports run in the browser, so speed and maximum length depend on your computer.